

Volatility and Under-Insurance in Economies with Limited Pledgeability: Evidence from a Frost Shock

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Motivation

- Under-insurance is prevalent in many settings with financially constrained households and small firms, even when premiums are subsidized
 - e.g. health, home, flood, wildfire, **crop**
- Many papers have sought behavioral explanations, but what if financial frictions result in under-insurance being privately optimal?
- What are the equilibrium consequences of under-insurance?
 - Prices, Volatility, Investment.
- How policy effectiveness depends on this channel, in particular with higher climate risks?

Research Question

In response to an unexpected net worth shock, commonly seen in environmental settings, how does the interaction of insurance and financial constraints affect firms' subsequent demand for credit/hedging, and how does investment react?

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Our Contribution

- **Model:** Limited pledgeability generates endogenous under-insurance and differential responses to the same shock by insurance status; GE amplification through the coffee price
- **Empirics:** Lasting differential responses from a one-time shock. Uninsured have, for each pct pt damage in capital stock:
 - 1.4 pct pt less outstanding debt (0.4 pct pt immediate; 1.4 pct pt long-run)
 - 20 months longer maturity (renegotiated, not new debt)
 - 0.38% lower investment vs 0.24% increase for insured
- **Policy:** Emergency credit lines are ineffective when the same collateral constraint limits both credit and insurance. Loosening the borrowing constraint (e.g., via CPR futures, better land registries) is more effective.

Comparison to the Literature

Causes and consequences of under-insurance: Caballero and Krishnamurthy 2003, Dávila and Korinek 2018, Rampini and Viswanathan 2022, etc.

- **Our model predicts that limited pledgeability of collateral results in coffee farmers with low net worth neither saving to use internal funds nor hedging against negative shocks, with GE consequences on the coffee price and distribution of farmer wealth.**

Credit and insurance: Cole and Xiong 2017; Casaburi and Willis 2018; McIntosh, Sarris, and Papadopoulos 2013; Annan 2022; Lane 2024, etc.

- **We show that credit and insurance behave as complements, from pledgeability constraints that bind after a shock rather than ex ante.**

Financial transmission of weather shocks: Bergman, Iyer, and Thakor 2020; Brown, Gustafson, and Ivanov 2021; Cortés and Strahan 2017, etc.

- **Our setting and mechanism differ from each of these papers: the physical shock, causing a reduction in the capital stock, interacted with insurance is what generates our financial and hedging results. We also have detailed payment and credit data for a granular exploration.**

Currently in the Draft: Partial Equilibrium

- Risk-averse firms are subject to shocks to their capital stock
- Firms can write contracts with risk-neutral financial intermediaries conditioning on state (shock) realization under limited pledgeability
- **Endogenously Incomplete Markets:** Equivalence between
 - the constrained optimal contract maximizing farmers' utility
 - a market featuring one-period state-contingent savings (insurance) and non-state-contingent debt contracts, all with collateral constraints

Theory - Overview

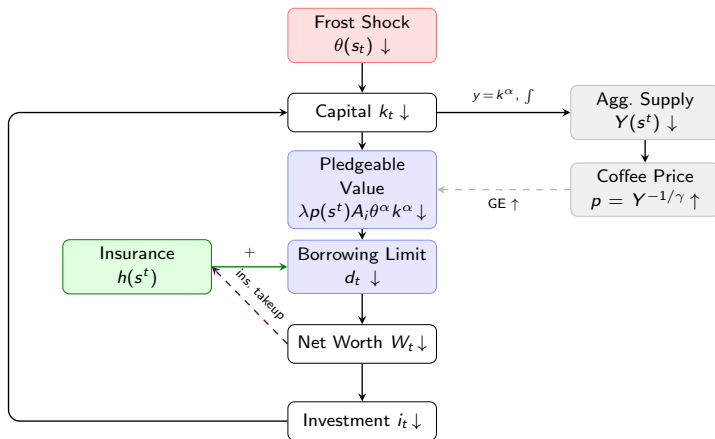
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Today: Preliminary Discussion of General Equilibrium with Endogenous Coffee Price

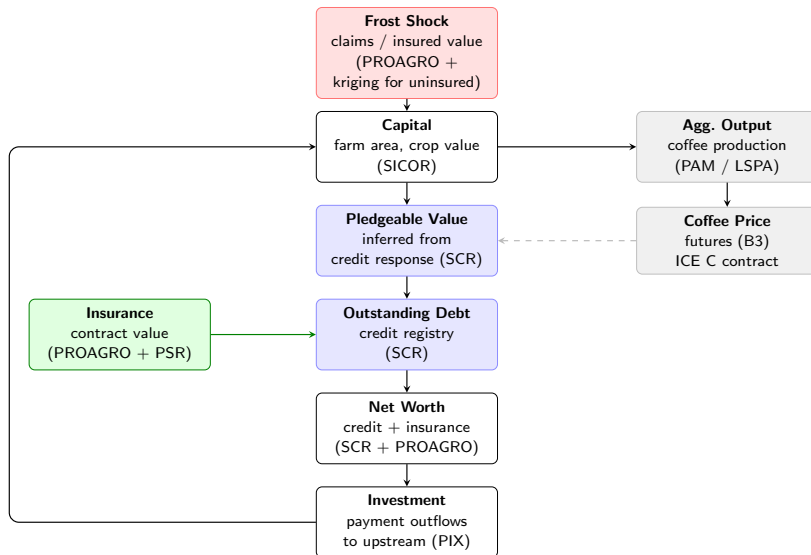
- Debt limit depends on the coffee price, determined by the aggregate state, as well as the idiosyncratic state
- Work-In-Progress: Estimation with SMM using the empirical results.

Model: Mechanism Overview



Loop (left): $k_i(s_t) = \theta_i(s_t)(i_{i,t-1} + k_{i,t-1})$. Dashed gray: GE price feedback. **Green**: insurance expands borrowing limit. **Violet dashed**: net worth governs future insurance takeup.

Data Counterpart to Each Model Component



Risk-averse farmers of type a maximize stochastic utility over an infinite horizon:

$$U_t^E(\{c_t\}) = \mathbb{E} \left[\sum_{\ell=0}^{\infty} \beta^{\ell} u(c_{t+\ell}) \right],$$

Output from capital:

$$y_{it} = A_a k_{it}^{\alpha},$$

Capital is subject to an idiosyncratic frost shock θ :

$$k_i(s_t) = \theta_i(s_t) (i_{i,t-1} + k_{i,t-1}), \quad \theta_i(s_t) \in \{0.3, 1 - \delta\}.$$

Limited Commitment: Farmers can default and keep fraction $(1 - \lambda)$ of $p_t A_a k_{it}^{\alpha}$; cannot be excluded from credit markets.

Optimal Contract: Farmer Problem

For a farmer of type a , choose $\{c(s^t), t(s^t), k(s^t)\}_{t \geq \tau}$ to maximize

$$U(\{c(s^t)\} \mid s^\tau) := E_\tau \left[\sum_{\ell=0}^{\infty} \beta^\ell u(c(s^{\tau+\ell})) \right].$$

Subject to the budget constraint, for all $t \geq \tau$:

$$\underbrace{\tilde{p}(s^t) A_a \theta(s^t)^\alpha k(s^{t-1})^\alpha - t(s^t)}_{\text{Net worth, } W(s^t)} \geq c(s^t) + k(s^t).$$

Endogenously Incomplete Markets

Proposition: A consumption allocation is the outcome of the optimal contract if and only if the allocation is the outcome of an economy where farmers only have access to a sequence of one-period state-contingent savings contracts $\{h(s^t)\}_{t \geq \tau}$ and one period non-state-contingent debt contracts $\{d_t\}_{t \geq \tau}$:

$$\underbrace{d_t}_{\text{Debt}} \leq \underbrace{\lambda p(s^t) A_a \theta(s_t)^\alpha k(s^t)^\alpha}_{\text{Pledgeability constraint}} + \underbrace{h(s^t)}_{\text{Insurance}}, \quad \forall s^t$$

$$d_t \geq 0, \quad h(s^t) \geq 0, \quad \forall s^t$$

Intuition

- ① Borrowing more than what is pledgeable requires insurance.
- ② But insurance premium needs to be paid **up front**.

Implication: credit and insurance act as **complements**, not substitutes.

Recursive Problem

Defining $b(s') := h(s') - d$, farmers of type a solve:

$$V_a(W, s) = \max_{c, k, \{W(s'), b(s')\}} u(c) + \beta \mathbb{E}[V_a(W(s'), s') \mid s]$$

$$\text{s.t.} \quad W + \sum_{s'} \pi(s'|s) \frac{b(s')}{R} \geq c + k,$$

$$\tilde{p}(s') A_a k^\alpha + \theta(s') k - b(s') \geq W(s'), \quad \forall s',$$

$$\lambda \tilde{p}(s') A_a \theta(s')^\alpha k^\alpha \geq b(s'), \quad \forall s'.$$

- $b(s') > 0$: net **insurance** payout in state s' ; $b(s') < 0$: net **debt** repayment.
- The collateral constraint pins **insurance** and **debt** *jointly*, as **complements**.

Price determination: Coffee price from CES demand:

$$p(s^t) = Y(s^t)^{-\frac{1}{\gamma}},$$

where $\gamma > 0$ is the demand elasticity and $Y(s^t)$ is aggregate supply:

$$Y(s^t) = \int_0^1 \theta_i(s_t)^\alpha A_i k_{it}^\alpha di$$

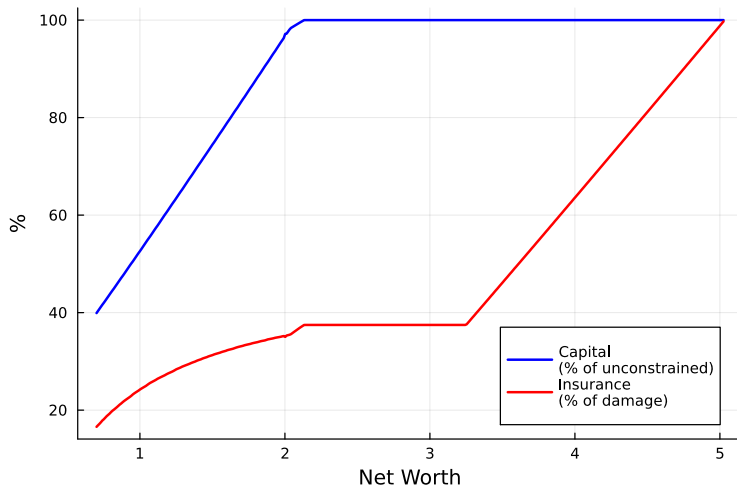
We set $\gamma = 0.4$ (inelastic demand, consistent with commodity markets). Interest rate $R > 1$ is exogenous and constant.

GE channel: Widespread frost $\Rightarrow Y \downarrow \Rightarrow p \uparrow \Rightarrow$ pledgeable value $\uparrow$ (partial offset for tightened borrowing constraint)

GE Calibration

- Continuum mass 1 of farmers.
- Idiosyncratic shock: $\theta = 0.3$ (frost damage) or $\theta = 1 - \delta$ (normal depreciation/no frost damage).
- Four aggregate states combine the coffee price and the probability of damage:
 - ① $p(s_1) = 1, \quad \mathbb{P}(\theta_i(s_1) = 0.3) = s_n$
 - ② $p(s_2) = 1.5, \quad \mathbb{P}(\theta_i(s_2) = 0.3) = s_n$
 - ③ $p(s_3) = 1, \quad \mathbb{P}(\theta_i(s_3) = 0.3) = s_f$
 - ④ $p(s_4) = 1.5, \quad \mathbb{P}(\theta_i(s_4) = 0.3) = s_f, \quad s_f > s_n$
- Unconditional state distribution: $P_S = [0.1; 0.1; 0.6; 0.2]$.
- $\alpha = 0.3, \lambda = 0.4, \beta = 0.85, R = 1.1, \sigma = 2, \delta = 0.05$.
- Two farmer types, insured (A_H) and uninsured (A_L); the steady-state distribution of insurance is chosen to match the data.
- **Solution:** Krusell-Smith with endogenous grid method (EGM).

Model Implication: Insurance Increases with Net Worth



Model: Event Study Setup

We feed the model the following sequence of aggregate states:

$$S = [1, \underbrace{4}_{\text{Capital Shock}}, \underbrace{2, 2}_{\text{High Price}}].$$

- There exists a net-worth threshold $\underline{W}$: farmers starting with $W \leq \underline{W}$ are uninsured in period 2 (less than 20% damage covered).
- Initial net worth: uniform on $[0, \underline{W}]$ (uninsured) and $[\underline{W}, \overline{W}]$ (insured).
- Half of each group is randomly selected to receive the capital shock.

Model: TWFE Specification

We run the following TWFE on the simulated panel:

$$y_{it} = \sum_{\tau=-1}^2 \beta_{\tau}^{\text{I}} \text{Shock}_{i\tau} \text{Ins}_i + \sum_{\tau=-1}^2 \beta_{\tau}^{\text{NI}} \text{Shock}_{i\tau} (1 - \text{Ins}_i) + \alpha_i + \alpha_{gt} + \epsilon_{it}.$$

- $\text{Shock}_{i\tau}$: indicator for receiving the capital shock in period τ .
- Ins_i : insured status. β_{τ}^{I} : insured response. β_{τ}^{NI} : uninsured response.
- g : insurance group; cumulative effects plotted, pre-period normalized to 0.

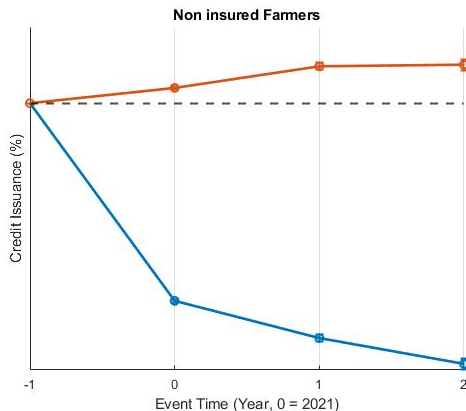
Non-Insured Summary

Debt Responses

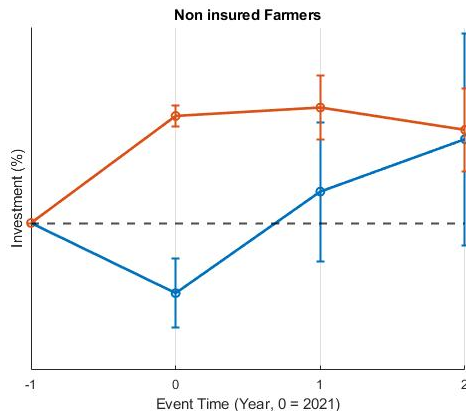
Investment Responses

Model-Implied Responses: Non-Insured Farmers

Non-Insured - Credit Responses



Non-Insured - Investment Responses



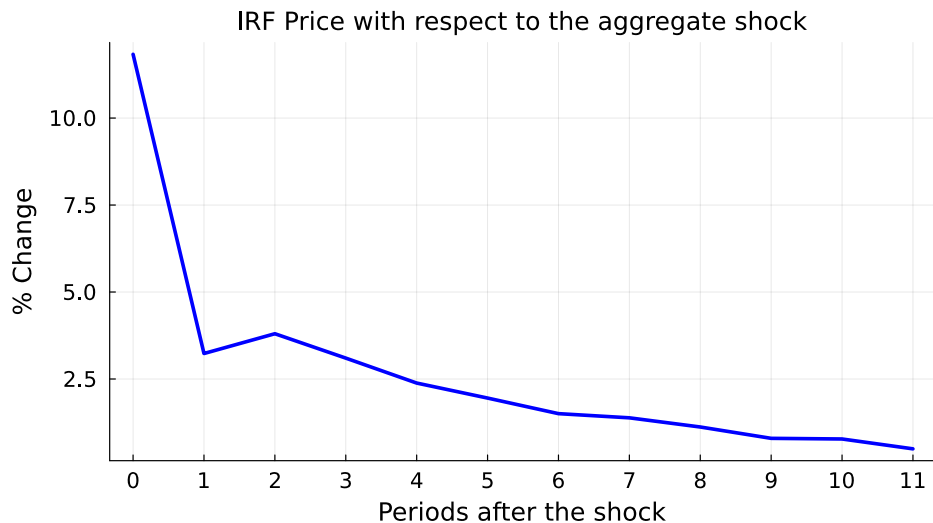
Non-insured reduce debt persistently; slow recovery on investment.

[Return to Model TWFE](#)

Insured vs Non-Insured Debt

Insured vs Non-Insured Investment

Endogenously Persistent Impact through the Coffee Price



Empirical Setting: Frost Shock in Brazil

- In July 2021, four frost waves struck agricultural regions in Brazil, with heterogeneous impact across crops and plots
- As Brazil accounts for almost 40% of the world's production, prices soared in 2021 and remained high during 2022.

Climate Capital Agricultural commodities + Add to myFT

Weather shocks in Brazil ripple across global commodities markets

FT

Coffee Prices Jump to Six-Year High as Brazilian Frost Threatens Crop

Arabica-bean prices have leapt almost a third in July in latest instance of extreme weather sending jitters through commodity markets

WSJ

Price Time Series

Futures Prices

Futures Volume

Map of Coffee Production

Empirical Setting: Coffee in Brazil

Coffee plants are perennial, so the frost negatively affected farmers' net worth



Coffee Price Time Series

Map of Coffee Production

All accessed through the Central Bank of Brazil:

- Publicly subsidized **rural credit and insurance** (PROAGRO) at the contract level.
- **Private agricultural insurance** at the contract level.
 - Also, OTC data at the contract level, with information about FX hedging and coffee future contracts.
- **Credit registry** (SCR) at the loan level.
 - We also use the **Rural Credit Registry** (SICOR), which also contains information about the farmer's activity (which crops the farm grows), farm area, partial data on expected and realized production, and geographic location of the farms.
- **Electronic payments** at the transaction level, with information about each counterparty's municipality and 7-digit CNAE sector (roughly the same as a 6-digit HS code)

Measurement

Summary Stats

Data Construction

Measuring the Frost Shock using Insurance Claims Data

- We observe contract-level insurance claims, identified at the farm level (longitude, latitude coordinates, and crop).
- We look at insured coffee farmers who did not claim for other reasons during the frost shock period.
- For a given farmer i in the above group, we observe farm coordinates x_i and ratio of frost claim value to insured value, fs_i that we extrapolate to uninsured farmers:

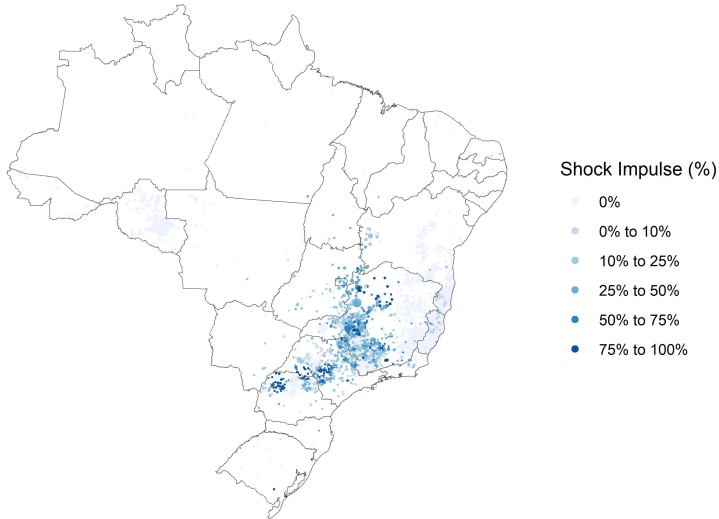
$$fs_j = \frac{\sum_{i \in \mathcal{N}_j} w_i(x_j) fs_i}{\sum_{i \in \mathcal{N}_j} w_i(x_j)}, \quad w_i(x_j) = \frac{1}{d(x_i, x_j)^\beta},$$

where $d(x_i, x_j)$ is the distance between farms, and $\mathcal{N}_j$ is the set of j k -nearest neighbors.

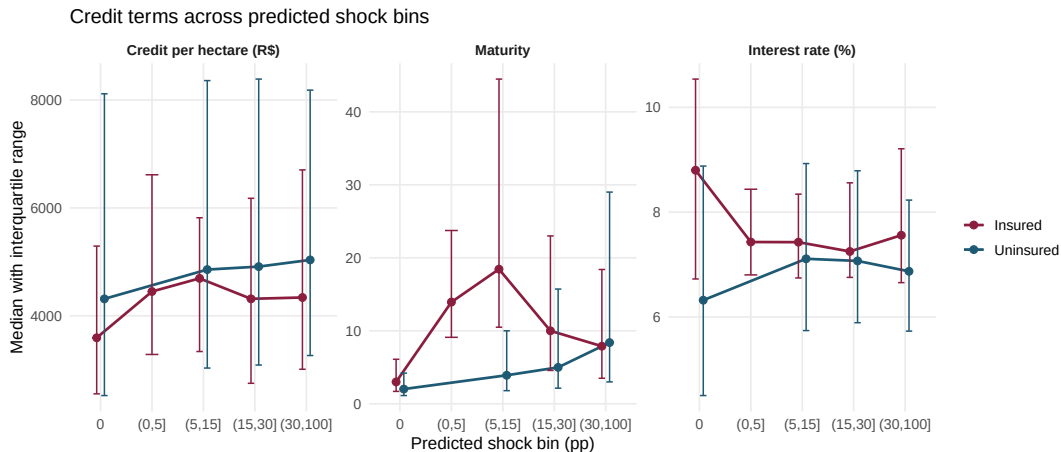
- Both β and k are chosen by cross-validation.

Frost Shock Measurement

Frost shock impulse to coffee, % of largest value



Credit Terms by Shock Susceptibility



Raw medians within each insurance group. The event study formalizes this with FEs and time dynamics.

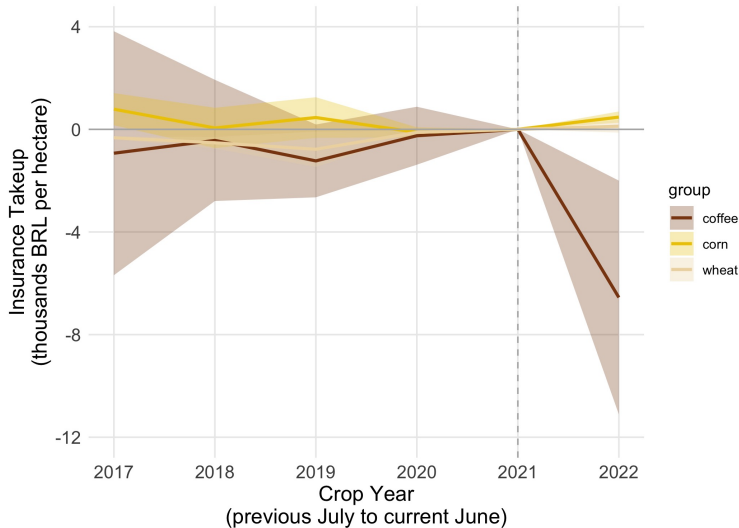
Insurance Regression Specification

$$y_{ijct} = \sum_{\tau=2017}^{2022} \beta_{\tau} \text{fs}_{ijc\tau} \text{Ins}_{ijc} + \alpha_{ijc} + \alpha_{jct} + \epsilon_{ijct}.$$

- i farmer, j municipality, c crop, t year (growing season).
- Ins_{ijc} : insured at the time of the shock.
- y_{ijct} : Insured Value (BRL) / Area (ha).
- **Intuition:** within municipality, insurance take-up of affected vs unaffected farmers, controlling for individual FEs.

Insurance prices unchanged

Affected Farmers Reduce Insurance Take-Up



Unaffected take up more

FX Hedging

Credit Regression Specification

We use the following specification for the credit event study (quarterly frequency).

$$y_{ijt} = \sum_{\tau=2019Q1}^{2023Q4} \beta_{\tau}^I fs_{ij\tau} Ins_{ij} + \sum_{\tau=2019Q1}^{2023Q4} \beta_{\tau}^{NI} fs_{ij\tau} No_Ins_{ij} + \alpha_{ij} + \alpha_{jgt} + \epsilon_{ijt}.$$

- i farmer, j municipality, t quarter, g insurance group.
- y_{ijt} : log outstanding debt.
- $fs_{ij\tau}$: frost shock severity (pct capital damaged) $\times \mathbf{1}[t = \tau]$.
- α_{ij} : farmer $\times$ municipality FE; α_{jgt} : municipality $\times$ group $\times$ time FE.
- **Intuition:** comparing credit to affected vs unaffected farmers within the same municipality, absorbing common time shocks by insurance status.

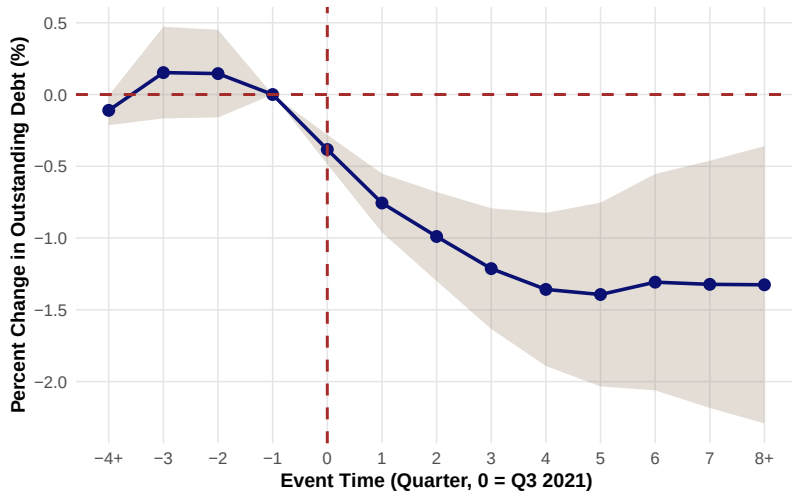
Non-Insured Debt Load

Non-Insured Debt Maturity

Insured Debt Load

Insured Debt Maturity

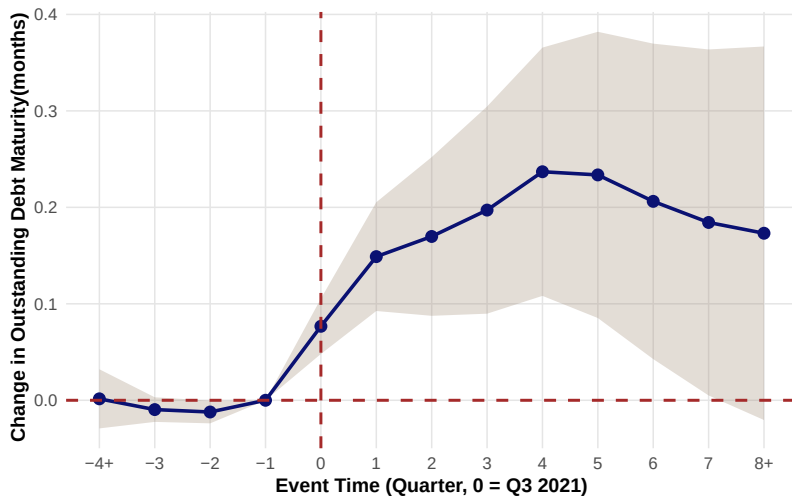
Non-Insured Farmers' Debt Load ($\hat{\beta}_{\tau}^{\text{NI}}$ Cumulative Effects)



Comparison with Insured

Regression

Non-Insured Farmers' Debt Maturity ($\hat{\beta}_{\tau}^{\text{NI}}$ Cumulative Effects)



Comparison with Insured

Regression

Payment Regression Specification

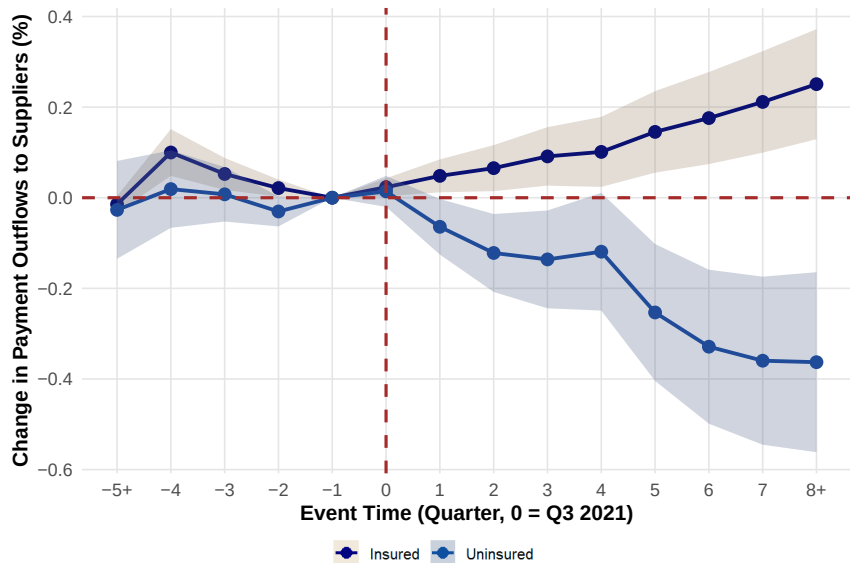
We use the following specification to proxy for investment using payments data aggregated quarterly:

$$y_{ijst} = \sum_{\tau=2019Q1}^{2023Q4} \beta_{\tau}^I fs_{ij\tau} Ins_i + \sum_{\tau=2019Q1}^{2023Q4} \beta_{\tau}^{NI} fs_{ij\tau} No_Ins_i \\ + \alpha_{is} + \alpha_{jmt} + \epsilon_{ijst}.$$

- i farmer, j municipality, s supplier, t quarter, m supplier 7-digit CNAE.
- y_{ijst} : log payment outflows to upstream suppliers (seeds, fertilizers, capital replenishment).
- α_{is} : farmer $\times$ supplier FE; α_{jmt} : municipality $\times$ CNAE $\times$ time FE.
- **Intuition:** within municipality, comparing payment outflows to a given sector from affected vs unaffected farmers, absorbing supplier-time shocks.

Skip to Conclusion

Insured $\hat{\beta}_T^I$ Invested More, Non-Insured $\hat{\beta}_T^{NI}$ Invested Less



Additional Results

- No significant effects on the interest rate nor *default rate*.
- Margin of adjustment was debt renegotiation, extending the maturity of existing debt.
- No significant effects on insured farmers' debt composition.
- Unaffected coffee farmers increased insurance take-up; annual crop farmers unchanged.

Default Results

Interest Rate Results

Skip to Conclusion

Implication: Emergency Credit Lines

- Emergency credit lines *intermediated* by the financial sector **may not be effective** when the intermediaries are unconstrained
- Corroborated in the data: the majority of what was allocated for emergency credit lines was not lent, and was lent disproportionately more to insured farmers.
 - Empirically, this pattern was also observed in the Covid-19 emergency credit lines in the US (Joaquim and Wang 2022).
- **Conclusion:** Emergency credit lines do not substitute for insurance when the same underlying friction limits both insurance and credit take-up.

Implication: the other side of climate stress testing

- Climate stress tests are designed to measure and understand the exposure of the financial sector to climate risk.
- This paper argues that in some settings, the exposure of the financial sector is actually **inefficiently low**.
- Actions that lead to less exposure may imply the real sector is the one absorbing the risk in the end.
- **GE Amplification:** In inelastic markets, consumers bear a substantial portion of the burden during periods of price volatility.
- **For policy:** these initiatives should be accompanied by policies that target the pledgability of assets.
 - Examples already happening: innovations in CPR futures, green credit bureaus, better land registries, and tokenized land.

Implication: Increase the Effective Pledgeability of Collateral

CPR (crop-settled) and CPR-F (money-settled) futures:

- Lenders can use satellite imagery to verify farmers' crop conditions and real-time harvest status
- By credibly pledging future harvest to lenders, farmers can increase borrowing capacity given a fixed amount of assets as collateral

Fractionalization from tokenization of land:

- Comprehensive registry of land + tokenization could allow farmers to pledge the unused value of land as collateral to other lenders, increasing competition between lenders in financing terms

Conclusion

- **Setting:** July 2021 frost shock to Brazilian coffee farmers provides a quasi-experiment in the propagation of a net-worth shock through financial and insurance markets
- **Model:** Limited pledgeability $\Rightarrow$ credit and insurance are complements, not substitutes. GE amplification through the coffee price generates persistence.
- **Empirics:** For each pct pt damage:
 - Insurance take-up falls (affected farmers), while unaffected farmers buy more
 - Uninsured borrow 1.4 pct pt less and extend maturity 20 months; insured borrow more
 - Uninsured cut upstream investment 0.38 pct pt; insured increase it 0.24 pct pt
- **Policy:** Emergency credit lines might miss the binding constraint. Potentially effective interventions directly increase pledgeability (CPR futures, tokenized land).
- **Next:** SMM estimation to quantify under-insurance and welfare counterfactuals

Appendix



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Lane, Gregory (2024). "Adapting to Climate Risk with Guaranteed Credit: Evidence from Bangladesh". In: *Econometrica* 92.2, pp. 355–386.



McIntosh, Craig, Alexander Sarris, and Fotis Papadopoulos (2013). *Productivity, Credit, Risk, and the Demand for Weather Index Insurance in Smallholder Agriculture in Ethiopia*. Tech. rep., pp. 399–417.



Rampini, Adriano and S. Viswanathan (2022). *Financing Insurance*. Tech. rep.

Numerical Solution

- Perceived Law of Motion (PLM): Given moments m_t (e.g. aggregate capital) of the distribution of $A_i k_{it}^\alpha$, farmers predict:

$$Y_{t+1}(S_{t+1}) = \int_0^1 \theta_i(s_{t+1})^\alpha A_i k_{i,t+1}^\alpha di = f(m_t, S_t, a),$$

where $f(\cdot)$ is a known functional form and a is a finite-dimensional parameter vector.

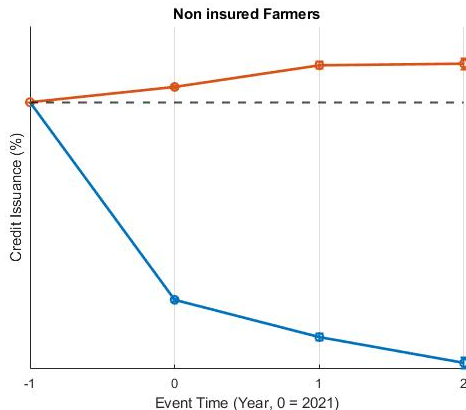
- Start with initial guess a_0 :
 - 1 Solve household problem for given a .
 - 2 Simulate economy, generate $\{m_t, S_t, Y_{t+1}\}$.
 - 3 Re-estimate a from simulated data.
- Repeat until change in a (or prediction error) is below tolerance

[Additional Details](#)

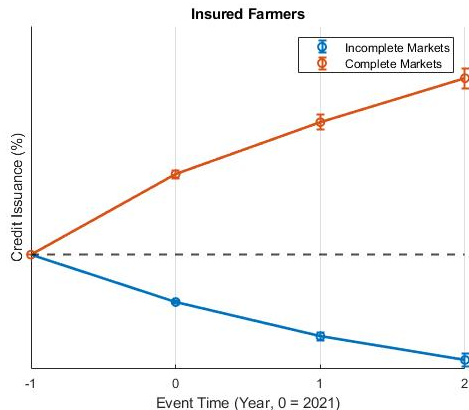
[Return to Calibration](#)

Model-Implied Debt Responses: Insured vs Non-Insured

Non-Insured ($\hat{\beta}_{\tau}^{\text{NI}}$)



Insured ($\hat{\beta}_{\tau}^{\text{I}}$)



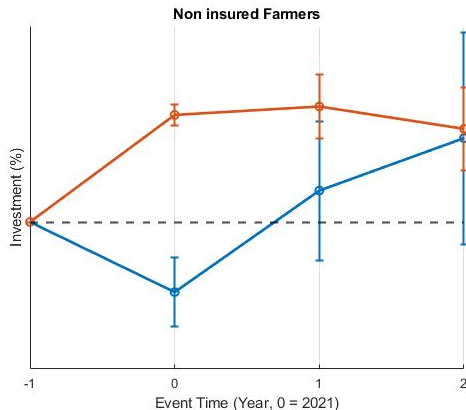
Non-insured reduce debt persistently; insured expand it to finance replanting.

[Return to Model TWFE](#)

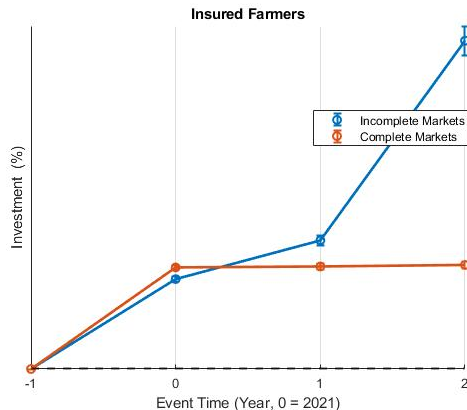
Investment Responses

Model-Implied Investment Responses: Insured vs Non-Insured

Non-Insured ($\hat{\beta}_\tau^{\text{NI}}$)



Insured ($\hat{\beta}_\tau^{\text{I}}$)



Insured replenish capital; non-insured cut investment, consistent with binding pledgeability constraint.

Return to Model TWFE

Debt Responses

Household Problem Solution via Endogenous Grid Method (EGM)

- For each aggregate price P and agent type a :
 - ① Construct $\tilde{\theta}_{j,s}$ from PLM parameters and current P .
 - ② Initialize V_w with marginal utility guesses.
 - ③ Consider cases on a fine wealth grid:
 - **Unconstrained:** Invert Euler equation to find c and w' for each future state.
 - **Partially constrained:** Some future-state borrowing limits bind.
 - **Fully constrained:** All future-state borrowing limits bind.
 - ④ Iterate until V_w converges; store policies $k(w, s)$, $w'(w, s, s')$, $c(w, s)$.
 - ⑤ Wrap policies into interpolation objects indexed by P .

[Return to Numerical Solution Summary](#)

Krusell-Smith Simulation and PLM Update

- **Simulation:**

- ① Simulate T periods with N agents (insured & uninsured) under given PLM.
- ② Aggregate capital and compute price from production function and shock structure.
- ③ Use precomputed policy interpolants to evolve individual (w, s) .

- **PLM update:**

- ① Regress $\log P_{t+1}$ on current-state dummies, interactions with $\log P_t$, and next-state dummies.
- ② Map coefficients to new PLM parameters.
- ③ Repeat outer loop until mean-squared change in parameters is small.

[Return to Numerical Solution Summary](#)

Differences from Standard Krusell-Smith

- **Heterogeneous collateral constraints:** Borrowing limits depend on future-state damage/productivity $\tilde{\theta}$, farmer type A_a , and differ across insured and uninsured agents.
- **Four-state individual process:** Combines aggregate state (good/bad) with individual damage (good/bad) $\Rightarrow$ 4 discrete states instead of 2.
- **Two agent groups:** Solve policy functions separately for insured and uninsured households, with different productivity types A_H and A_L .
- **Price determination:** Output price P_t from CES demand with elasticity 0.4^{-1} , rather than Cobb–Douglas rental rate.
- **EGM with multiple constraint regimes:** Explicitly handles unconstrained, partially constrained, and fully constrained cases in value function iteration.
- **PLM functional form:** Regression includes next-period state dummies S_{t+1} in addition to current S_t and $\log P_t$.

Insurance and Hedging among Coffee Farmers

- Our sample consists of all coffee farms that ever borrowed using rural credit lines in the period of 2020-2023.
 - 100,848 farms; the majority of farmers only own one farm
- Lack of insurance is prevalent among coffee farms in our sample.
In 2021:
 - 20% had quantity-based insurance.
 - 0.28% had FX hedging.
 - 0.13% entered in price hedging in local markets.
- Insured farms
 - are larger (42.6 vs 23.9 ha; 105 vs 59 acres).
 - and borrow at cheaper rates (5.6% vs 9.5% annually)
 - related to an insurance premium
- Around 27% were hit by the frost shock.

Data Construction: Linking Geolocated Farms to BCB Microdata

Insurance:

- Geolocated farm- or plot-level contracts from **SICOR / PROAGRO** and **MAPA private insurance**
- Coordinates, crop, area, insured value, premium, and claim amount/cause
- Lets us build a **claim-based frost shock** instead of relying only on coarse weather exposure

Identifiers:

- **CNPJ farms**: direct merge to firm-side BCB datasets, aggregated to the economic-unit / **CNPJ-8** level
- **CPF farms**: use shareholder and Receita Federal crosswalks to map individual borrowers into the same economic-unit structure
- Key step: place **CPF and CNPJ farmers on the same debtor-side unit** before merging payments and credit

Outcomes:

- **SCR credit registry**: debt balances, maturity, rates, default
- **Payments**: PIX, boletos, transfers, supplier CNAE
- **Hedging**: futures / FX contracts registered through B3 / BCB

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Measuring the Frost Shock using Insurance Claims Data

- We observe contract-level insurance claims, identified at the coffee farm level (longitude and latitude coordinates).
- Insurance claims include information on the total amount paid and the most prevalent event (e.g., frost, drought, etc).
 - Insurance claims are only paid after an agronomist validates the extent of the damages.
- Farm-level frost shock measure constructed from insurance claims data using a regularized spatial interpolation
 - Perhaps less measurement error of the shock impulse than the typical method of using weather data, because small differences in elevation, soil, and temperature seem to correspond to large differences in claims

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Frost Shock Measurement

- For a given insured coffee farmer i we observe:
 - x_i : Farm coordinates.
 - Our frost shock measure fs_i : the ratio of frost damage insurance claims to total insured value.
- For an uninsured farmer j we compute the frost shock measure fs_j via a regularized spatial interpolation:

$$fs_j = \frac{\sum_{i \in \mathcal{N}_j} w_i(x_j) fs_i}{\sum_{i \in \mathcal{N}_j} w_i(x_j)}, \quad \text{weights } w_i(x_j) = \frac{1}{d(x_i, x_j)^\beta},$$

where $d(x_i, x_j)$ is the distance between farms, and $\mathcal{N}_j$ is the set of j k -nearest neighbors.

- Both β and k are chosen by cross-validation.

CDF of frost shock for insured vs uninsured

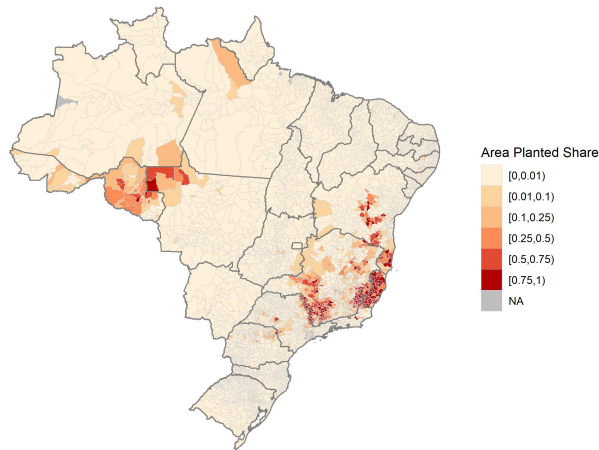
Frost Shock Measurement

Frost shock impulse to coffee, % of largest value



Coffee Planting Share

Coffee Area Planted Share of Agriculture in 2021, by Municipality



Source: Produção Agrícola Municipal

Frost shock measure: insurance claims vs weather

Left-hand-side is the ratio of total claims to insured value for farmer f growing crop c with insurer i in municipality j in year t . The right-hand-side is the frost shock defined as the crop-specific claims to insured value for crop year 2021.

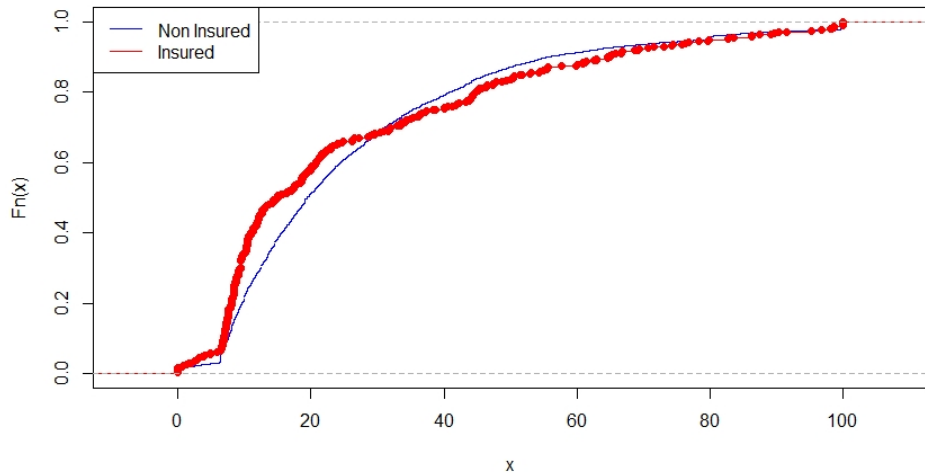
$$\left(\frac{\text{Claims}}{\text{Insured Value}} \right)_{fijt} = \sum_{\tau=2017}^{2021} \beta_{\tau} \text{fs}_{fcij} + \alpha_f + \alpha_{cijt} + \epsilon_{fcijt}.$$

Sanity Check: Close to 1-for-1 Impact on Insurance Claims for Frost

Table: Regressions of Total Claim-to-Insurance on Frost Claim-to-Insurance

Coffee shock 2021	0.87 [17.2]	0.76 [30.4]	0.76 [29.7]
Corn shock 2021	0.56 [42.6]	0.61 [80.5]	0.55 [69.5]
Wheat shock 2021	0.76 [76.6]	0.77 [102.6]	0.75 [93.5]
Farmer FE	X	X	X
Crop x Year FE	X	X	
Crop x Year x Insurer FE			X
Farm area control	X	X	X
Expected output control	X		
Observations	1,197,291	1,617,602	1,598,056

Frost Shock Measure: Insured vs Non-Insured



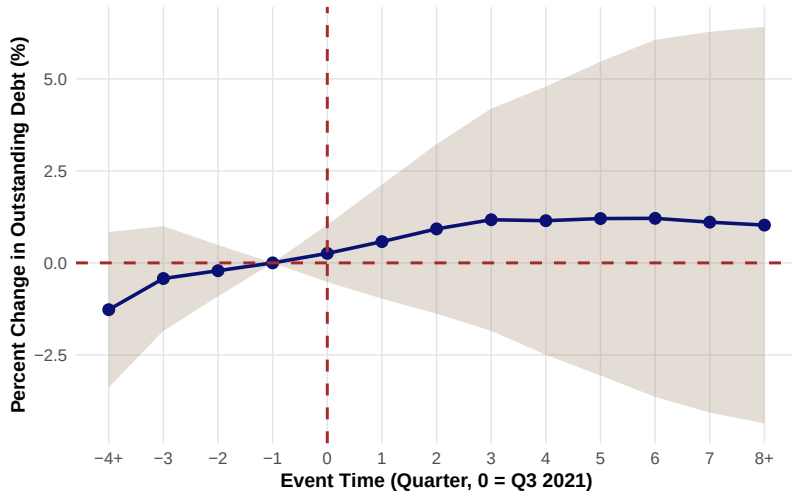
Robustness Check: Net Worth Shock

Table: Regressions of Pct Damage on Lagged Net Worth Shock and Shock Indicators

Lag Net Worth Shock	-43.2		
	[-47]		
Lag Shock Indicator 1		-1728.8	
		[-56]	
Lag Shock Indicator 2			-103.5
			[-5.3]
Lag Shock Indicator 3			-1664.4
			[-47.3]
Farmer x Crop FE	X	X	X
Crop x Year FE	X	X	X
Observations	750,348	750,348	750,348

Return

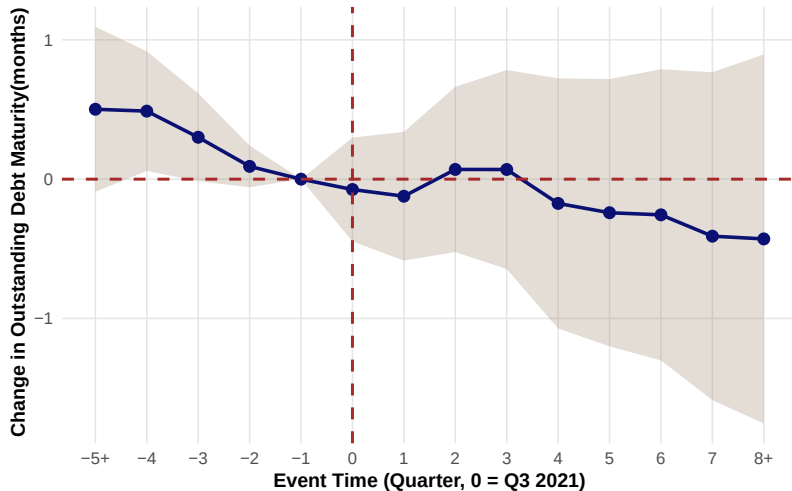
Insured Farmers' Debt Load ($\hat{\beta}_{\tau}^I$ Cumulative Effects)



Comparison with Non-Insured

Regression

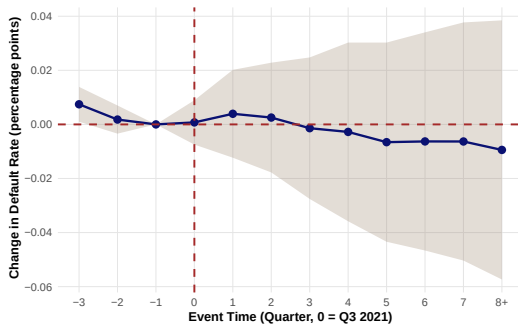
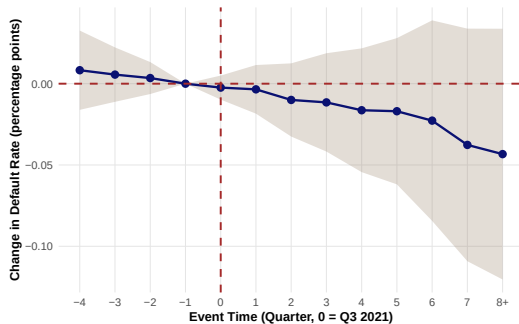
Insured Farmers' Debt Maturity ($\hat{\beta}_\tau^I$ Cumulative Effects)



Comparison with Non-Insured

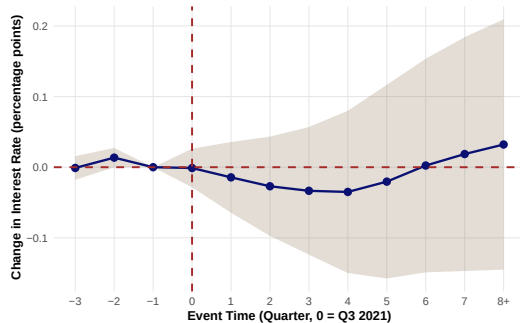
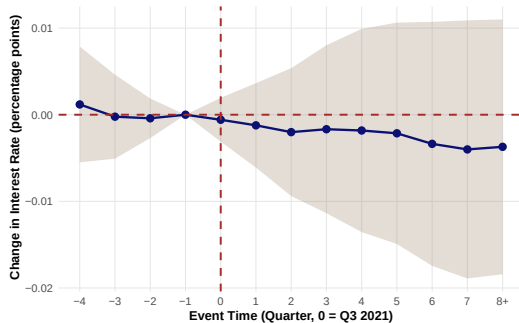
Regression

Default ($\hat{\beta}_{\tau}^I$ on left, $\hat{\beta}_{\tau}^{NI}$ on right)



[Return to Additional Results](#)

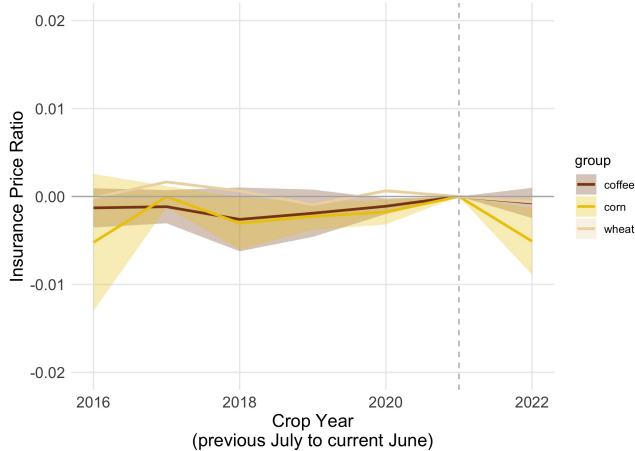
Interest Rate ($\hat{\beta}_{\tau}^I$ on left, $\hat{\beta}_{\tau}^{NI}$ on right)



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Impact on Insurance Prices

$$\left(\frac{\text{Premium}}{\text{Insured Value}}\right)_{fcijt} = \sum_{\tau=2017}^{2022} \beta_{\tau} dh_j + \alpha_f + \alpha_{cijt} + \epsilon_{fcijt}.$$



Impact on Insurance Prices

Table: Regressions of $\frac{\text{Premium}}{\text{Insured Value}}$

Coffee shock 2021	0 [0.4]	-0.001 [-0.8]	-0.003 [-1.8]
Corn shock 2021	-0.007 [-11.8]	-0.005 [-2.6]	0.006 [5.9]
Wheat shock 2021	-0.001 [-0.9]	-0.001 [-2.2]	0.002 [1.1]
Farmer FE	X	X	X
Crop x Year FE	X	X	
Crop x Year x Insurer FE			X
Farm area control	X	X	X
Expected output control	X		
Observations	1,840,382	2,336,587	2,362,363

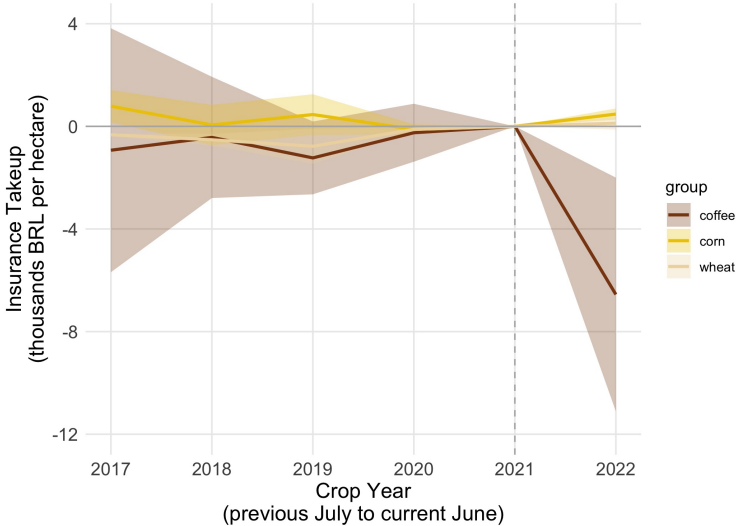
Insurance Regression Specification

$$y_{ijct} = \sum_{\tau=2017}^{2022} \beta_{\tau} \text{fs}_{ijc\tau} \text{Ins}_{ijc} + \alpha_{ijc} + \alpha_{jct} + \epsilon_{ijct}.$$

- i is farmer, j is municipality, c is crop, t is year (growing season)
- Ins_{ijc} is an indicator for insured at the time of the shock
- y_{ijct} is the response variable of interest: $\frac{\text{Insured Value (BRL)}}{\text{Area (Hectares)}}$.
- **Intuition:** comparing within the same municipality, insurance take-up of affected farmers versus unaffected farmers (after accounting for individual fixed effects).

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Affected Farmers Take up Less Quantity-Based Insurance



No impact on insurance prices

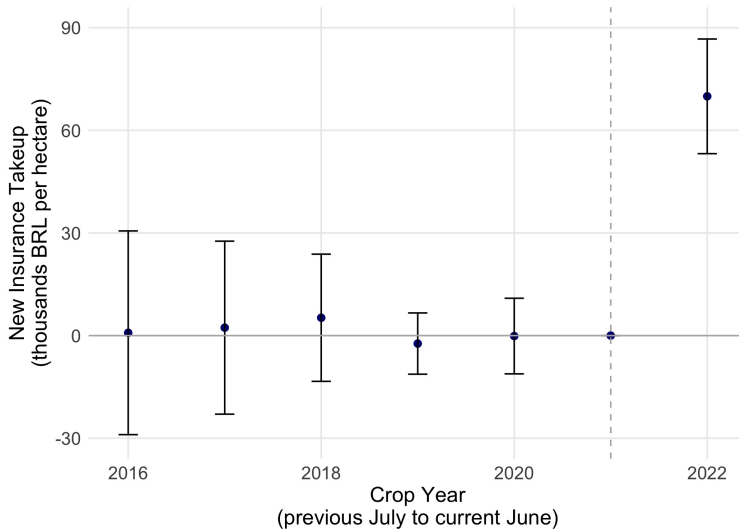
Unaffected farmers take up more

FX Hedging

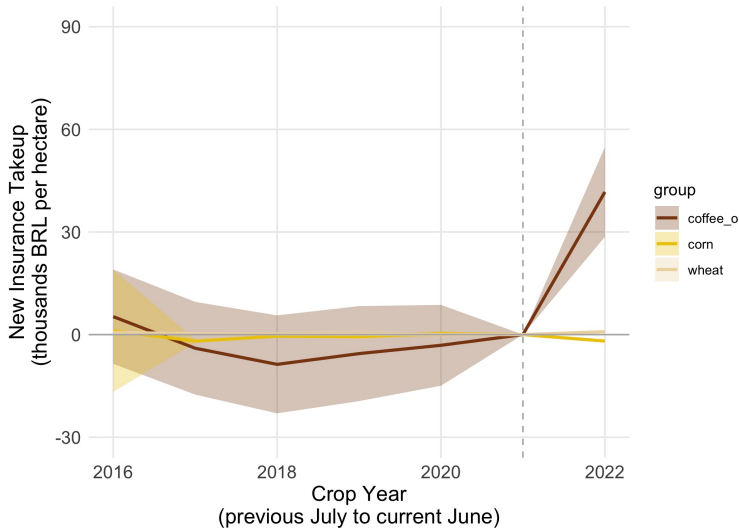
Return to Additional Results



Increase in New Insurance Take-Up by Municipality



Unaffected Farmers Purchase More Quantity-Based Insurance



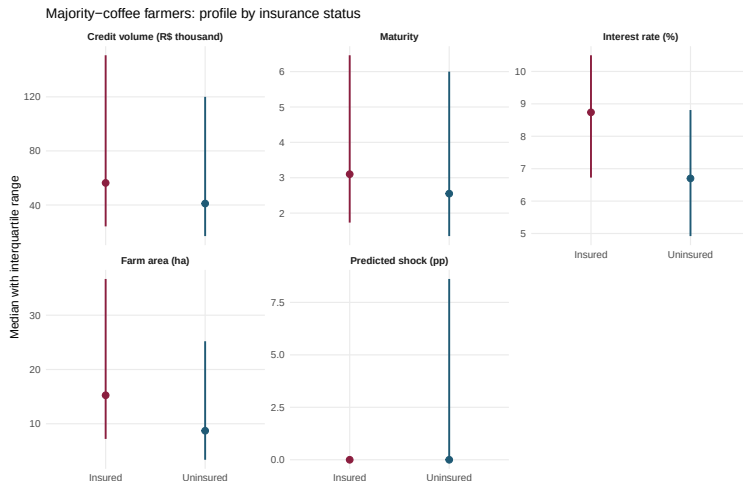
But Affected Coffee Farmers Take Up Less Insurance

Table: Regressions of $\frac{\text{Insured Value (BRL)}}{\text{Area (Hectares)}}$

Coffee shock CY 2022	-6552 [-2.8]	-9054 [-6.5]	-2867 [-2.5]
Corn shock CY 2022	477 [4.3]	-151 [-1.3]	217 [2.7]
Wheat shock CY 2022	121 [1]	105 [0.5]	78 [0.5]
Farmer FE	X	X	X
Crop x Year x Muni FE	X	X	
Crop x Year x Muni x Insurer FE			X
Expected output control	X		
Observations	1,840,378	2,359,865	2,334,632

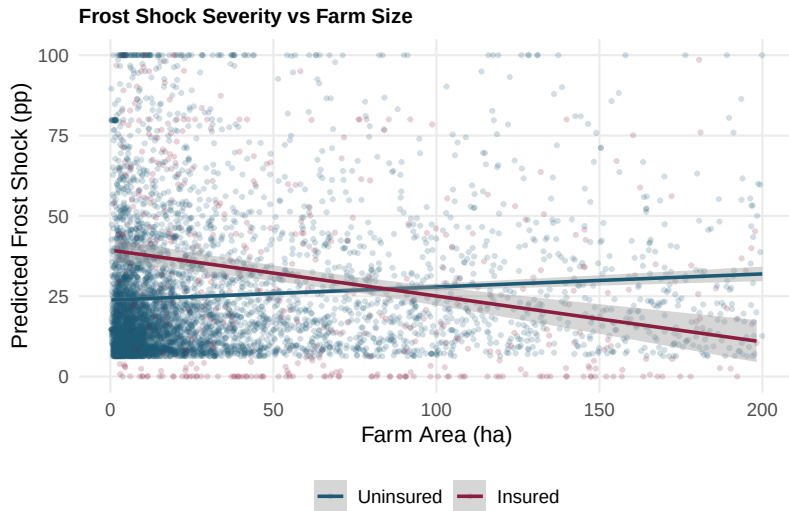
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Insured and Uninsured Farmers Differ Substantially Ex Ante



Selection prevents a simple insured vs. uninsured comparison — we instead compare *within* each group, more- vs. less-affected farmers, conditional on fixed effects. [Return](#)

Farm Size Does Not Predict Insurance Status

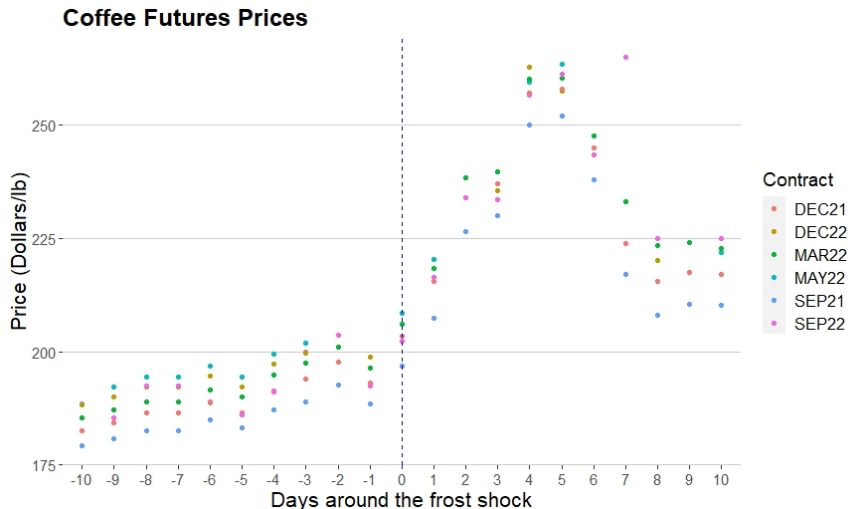


Coffee Prices

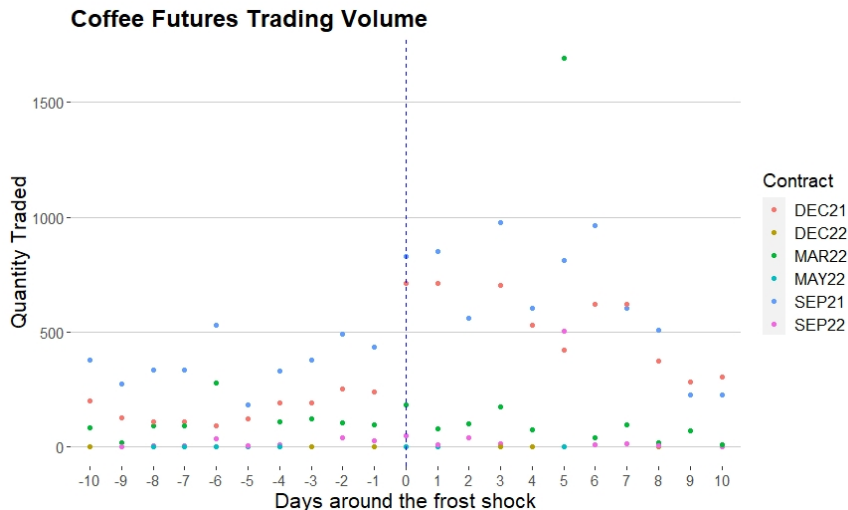


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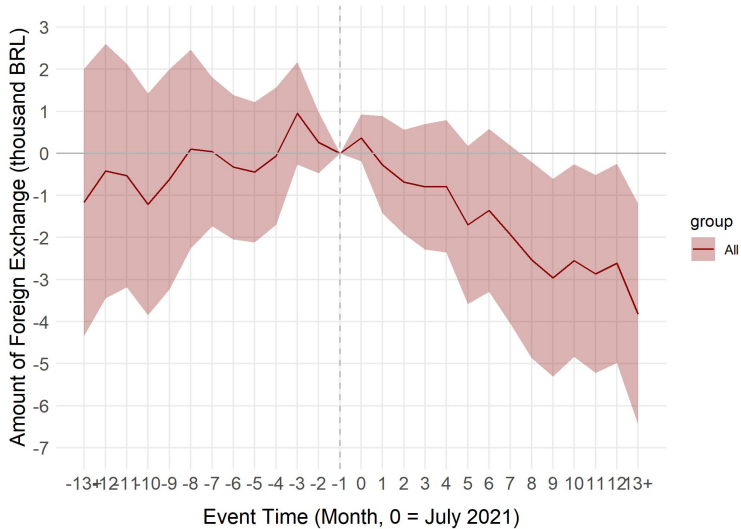
Futures Prices Around the Frost Shock



Trading Volume Around the Frost Shock



FX Hedging



Return